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Time-R1: Towards Comprehensive Temporal Reasoning in LLMs

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Abstract

Large Language Models (LLMs) demonstrate impressive capabilities but lack robust temporal intelligence, struggling to integrate reasoning about the past with predictions and plausible generations of the future. Meanwhile, existing methods typically target isolated temporal skills, such as question answering about past events or basic forecasting, and exhibit poor generalization, particularly when dealing with events beyond their knowledge cutoff or requiring creative foresight. To address these limitations, we introduce *Time-R1*, the first framework to endow a moderate-sized (3B-parameter) LLM with comprehensive temporal abilities: understanding, prediction, and creative generation. Our approach features a novel three-stage development path; the first two constitute a *reinforcement learning (RL) curriculum* driven by a meticulously designed dynamic rule-based reward system. This framework progressively builds (1) foundational temporal understanding and logical event-time mappings from historical data, (2) future event prediction skills for events beyond its knowledge cutoff, and finally (3) enables remarkable generalization to creative future scenario generation without any fine-tuning. Strikingly, experiments demonstrate that Time-R1 outperforms models over 200 times larger, including the state-of-the-art 671B DeepSeek-R1, on highly challenging future event prediction and creative scenario generation benchmarks. This work provides strong evidence that thoughtfully engineered, progressive RL fine-tuning allows smaller, efficient models to achieve superior temporal performance, offering a practical and scalable path towards truly time-aware AI. To foster further research, we also release *Time-Bench*, a large-scale multi-task temporal reasoning dataset derived from 10 years of news data, and our series of *Time-R1* checkpoints.[1]

1 Introduction

Large Language Models (LLMs) have achieved remarkable success across a spectrum of language understanding, generation, and even some complex reasoning tasks[1–3]. However, a persistent shortcoming in even the most advanced LLMs is their temporal reasoning ability[4, 5]. This encompasses several key capacities[6–8]: accurately interpreting temporal relationships within their existing knowledge base (such as inferring event times, time differences, event order, and completing temporal entities), predicting the timing of future events based on learned patterns, and creatively generating plausible future events anchored in time. Studies have shown that most LLMs indeed struggle to update or contextualize knowledge under time constraints [9]; even frontier models have been observed to perform worse than some smaller models in tasks that require integrating new temporal information [10]. This suggests a systemic weakness in how current LLMs grasp time. This weakness stems from multiple factors: architectural limitations [11], such as the lack of explicit

[1]Our code, the Time-Bench dataset, and Time-R1 model checkpoints are available at the project repository: https://github.com/ulab-uiuc/Time-R1 and via our Hugging Face Collection: https://huggingface.co/collections/ulab-ai/time-r1-682626aea47cb2b876285a16.

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module representation of time; the static nature of their training corpora [12], which inevitably become outdated; and the non-chronological training process [13], where temporal information across different periods is processed concurrently rather than sequentially, hindering the development of robust logical mappings between events and their corresponding times.

While existing research aims to enhance temporal reasoning—for instance, Zhao *et al.* [13] aligned LLM knowledge to target times, Kim *et al.* [9] improved temporal consistency, and Yuan *et al.* [5] focused on future event prediction, with other works exploring representation methods [14, 15]—these efforts often target isolated skills. They typically fall short of endowing LLMs with unified, comprehensive temporal intelligence that spans past understanding, future prediction, and creative, time-anchored generation, especially for events beyond their knowledge cutoffs [13, 5].

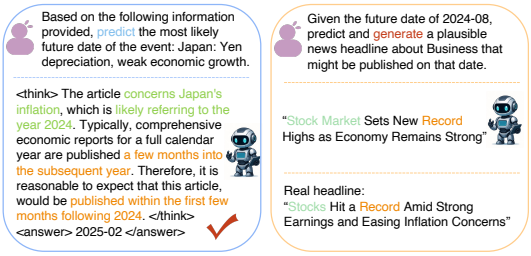

Figure 1: **Generated outputs from Time-R1 showcasing its capabilities.** (Left) Future Event Time Prediction (Stage 2). (Right) Creative Scenario Generation (Stage 3), with output compared to a real-world headline.

In this paper, we aim to bridge this gap by equipping a single 3B-parameter model with comprehensive temporal reasoning capabilities through multi-stage Reinforcement Learning (RL), which has become a powerful framework for improving LLM reasoning. Recent frontier models such as OpenAI-o1 [16] and DeepSeek-R1 [17] utilize RL methods like PPO [18] and GRPO [19], proving effectiveness to learn complex reasoning capabilities, such as mathematical problem solving and multi-step logical deduction. We build upon Qwen2.5-3B-Instruct, a moderate-sized LLM, and demonstrate that through specialized training it can surpass models over 200× larger (for instance, DeepSeek-R1, a 671B-parameter model) on highly challenging temporal prediction and generation tasks. We propose a three-stage framework with RL and dynamic rewards to progressively establish the model's unified temporal capabilities, spanning temporal logic, future prediction, and time-anchored scenario generation: **(1) Stage 1 - Comprehension**: RL fine-tune the model using pre-cutoff data from a cold start on four fundamental temporal tasks – timestamp inference, time-difference estimation, events ordering, and masked time entity completion – to develop powerful logical mappings between events and their corresponding times. **(2) Stage 2 - Prediction**: Further train the model to predict events occurring after knowledge cutoff, thereby teaching it to utilize general reasoning ability built in Stage 1 to extrapolate trends and anticipate future outcomes. **(3) Stage 3 - Generation**: Directly have the model generate logical future scenario without fine-tuning, leveraging the capabilities obtained from the first two stages.

Through this staged curriculum, the LLM thus progresses from comprehending known temporal facts to skillfully navigating the complexities of the future. This advanced training culminates in robust capabilities for both predicting future event timelines and creatively generating plausible scenarios for unseen future contexts—addressing significant limitations in how current AI handles such challenging forward-looking tasks. Illustrative examples of these advanced future-oriented skills, such as Time-R1's proficiency in forecasting event dates and generating contextually appropriate news headlines for future dates (as depicted in Figure 1), highlight the practical efficacy of our approach.

In summary, the key contributions of our work are as follows: **(1) Unified Temporal Reasoning in One Model:** We introduce the first LLM that exhibits a holistic temporal reasoning ability encompassing logic, prediction, and generation. **(2) Small Model, Big Performance:** We show that a relatively small 3B model, when fine-tuned with our meticulously designed multi-stage dynamic-reward RL strategy, can match or even exceed the performance of models with hundreds of billions of parameters (*e.g.*, the 671B-parameter R1 model) on temporal prediction and generation tasks. **(3) Fast Adaptability and Cost Efficiency:** Our approach demonstrates that temporal knowledge can be continuously refreshed in a cost-effective manner. A 3B model can be quickly fine-tuned on new data as time progresses, which is infeasible for a hundreds of billion model that would require enormous computational resources (on the order of millions of dollars for fine-tuning). **(4) Resources for the Community:** To encourage further research in temporal-aware AI, we release **Time-Bench**, a dataset of over 200,000 examples with explicit temporal annotations covering diverse tasks including timestamp inference, time-gap estimation, event ordering, and temporal entity completion. We also

release **Time-R1**, a series of high-performing and continuously updatable temporal reasoning model checkpoints, offering a strong foundation for future time-aware LLM development and iterative refinement.

2 Related Work

Temporal Reasoning in LLMs. While adept at many complex tasks [17, 20], LLMs struggle significantly with temporal reasoning—understanding time and event interrelations—a faculty crucial for comprehensive world understanding and interaction [4, 21, 6]. Recent studies increasingly target these deficiencies, often focusing on specific temporal facets. For example, some efforts aim to improve temporal accuracy by aligning LLM knowledge with a target time for time-sensitive questions [13]. Meantime, some investigate methods for better integrating temporal information into model representations [14], while others explore leveraging external knowledge sources or structured representations like temporal graphs to augment LLM capabilities [15]. However, LLMs exhibit particularly poor generalization when reasoning about the future, especially for events beyond their knowledge cutoff or tasks requiring creative foresight. Consequently, robust methods for direct, challenging future event prediction or creative scenario generation remain scarce in the literature. While some initiatives explore future event prediction and forecasting (e.g., Yuan *et al.* [5] employed instruction tuning to predict event occurrences from past contexts), comprehensive approaches addressing the full spectrum of complex and creative future-oriented reasoning are largely underdeveloped.

Reinforcement Learning in LLMs. Reinforcement learning (RL) has recently attracted attention due to its scalability and enhanced generalization capabilities. Building on policy optimization algorithms like PPO [18], reinforcement learning from human feedback (RLHF) — the first application of RL to large language models — has become a standard paradigm for aligning LLMs with desired behaviors [22, 23]. Recent advances aim to simplify or improve this process: Direct Preference Optimization (DPO) [24] and Simple Preference Optimization (SimPO) [25] replace the conventional RL loop with more direct optimization of preference-based rewards, eliminating the need for a separate reward model or reference policy. Other methods are tailored specifically for LLMs; for instance, Group Regularized Policy Optimization (GRPO) [19] introduces a group-based reward formulation in place of a single critic, achieving more stable training and better generalization. Likewise, Ahmadian *et al.* [26] revisit classic policy gradient techniques [27] to propose RLOO (REINFORCE-Leave-One-Out), an online RL algorithm that refines LLM policies with reduced variance and cost. These RL-driven approaches have demonstrated notable gains in LLM reasoning capabilities. In particular, GRPO and related strategies have yielded state-of-the-art performance on complex reasoning tasks including mathematical problem solving [19, 28], search engine interaction and knowledge retrieval [29, 30], code generation tasks [31] and others [32–34]. Despite these successes, the application of reinforcement learning to temporally-grounded reasoning remains underexplored. This gap suggests an opportunity to leverage RL methods to develop unified, time-sensitive reasoning abilities in future LLMs.

3 Method

This section details the **Time-R1** methodology for enhancing LLM temporal capabilities via Reinforcement Learning (RL) fine-tuning. We introduce a novel three-stage training framework (Section 3.2) guided by a dynamic, rule-based reward system (Section 3.3). We first outline the underlying RL optimization setup using Group Relative Policy Optimization (GRPO) (Section 3.1) before detailing these core framework and reward components.

3.1 Reinforcement Learning Fine-tuning for Temporal Reasoning

Our approach employs reinforcement learning (RL) to fine-tune a Large Language Model (LLM) for complex temporal reasoning tasks. The core process involves interaction between the LLM policy and a rule-based environment. Given a prompt x detailing a specific temporal task, the LLM, parameterized by θ, generates an output sequence y autoregressively according to its current policy $\pi_\theta(y \mid x) = \prod_{t=1}^{|y|} \pi_\theta(y_t \mid x, y_{<t})$.